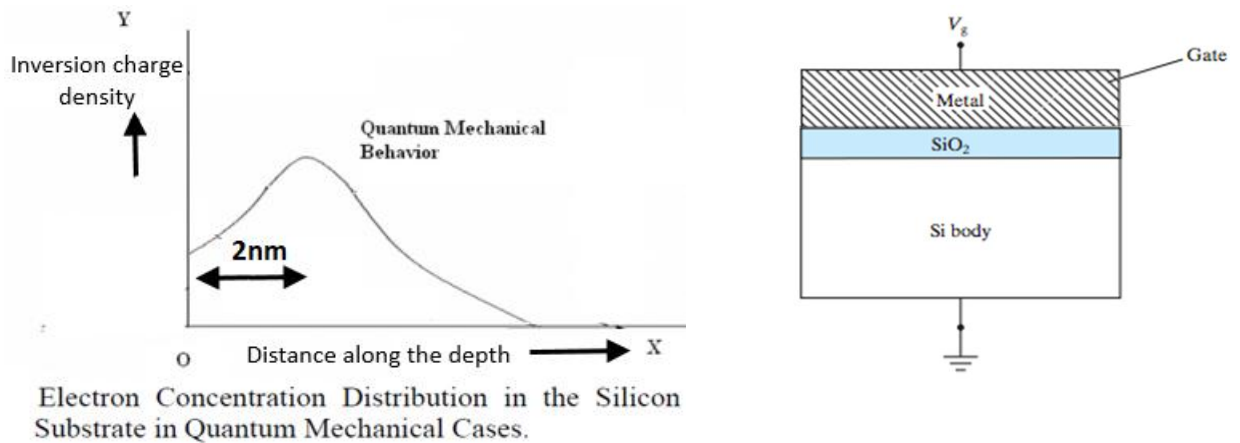
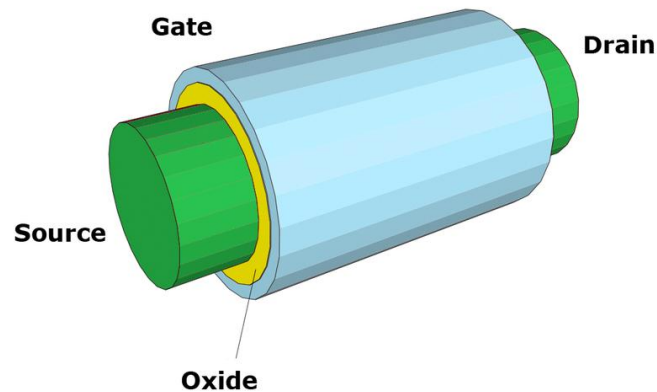


- Q1.** (a) Discuss the quantum confinement and discretization of energy band diagram in nanometer scale materials/devices.
- (b) The charge concentration at the SiO_2 -Silicon interface (in Silicon substrate side) in quantum mechanical condition is shown below.

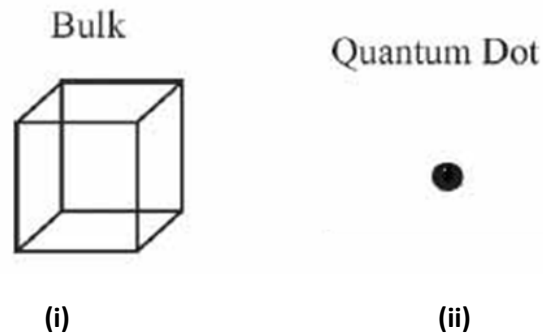


Consider Silicon nanowire structure as shown below.



Assume the silicon nanowire is operating under inversion mode. Draw the charge carrier density along the diameter (d) of the nanowire for; $d=10\text{nm}$ and $d=4\text{nm}$.

Q2 (a) Figures (i) and (ii) show bulk and quantum dot structure of the direct bandgap semiconductor, respectively. Write the relative energy band diagram of bulk and quantum dot and compare the energy band gaps (which bandgap is larger).



(b) Consider the below details of the quantum dot structure of the direct bandgap semiconductor (diameter 'd'). Given relation; $E=h \cdot f$ and $c=f \cdot \lambda$.

$$\text{Energy} = E = 3.9 \times 10^{-19} \text{ J}$$

$$h = \text{Plank's constant} = 6.626 \times 10^{-34} \text{ J.S}$$

$$C = \text{Speed of light} = 3 \times 10^8 \text{ m/s and } \lambda \text{ is wavelength in m.}$$

$$h \cdot c = 19.9 \times 10^{-26} \text{ J.m}$$

Find the colour of the light emitted when an electron moves from CB to VB of quantum dot. Refer to below visible light spectrum.

